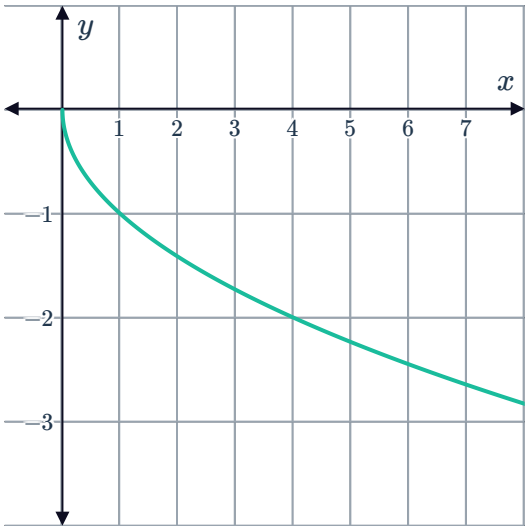
Graphs of square root functions

1 Consider the function $y = \sqrt{x}$.

- **a** Can y ever be negative?
- **b** As x gets larger and larger, what value does y approach?
- **c** Sketch the graph of $y = \sqrt{x}$.
- **d** Describe the rate of increase of the function as x increases.
- **e** State the axes intercepts.
- **f** Does the function have an asymptote?
- **g** Does the function have a limiting value?
- **h** Consider the function $y = 5\sqrt{x}$. How does this function differ from $y = \sqrt{x}$?

2 Consider the function $y = -\sqrt{x}$ graphed:

- **a** Can the function values ever be positive?
- **b** Describe the rate of decrease of the function as x increases.



3 Consider the function $y = \sqrt{x} + 3$.

- **a** Can y ever be negative?
- **b** As x gets larger and larger, what value does y approach?
- **c** Determine the y-intercept of the curve.
- **d** How many x-intercepts does it have?
- **e** Sketch the graph of the function.

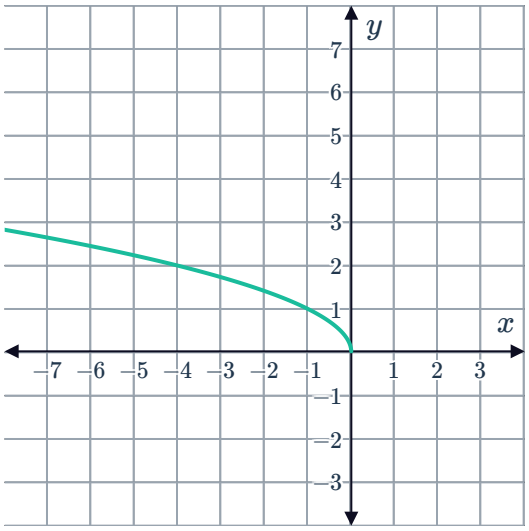
© ClipSAT: By Mr. Mohamed Abdallah +966597688647

4 Consider the graph of $y = \sqrt{-x}$:

a State the domain of the function.

b State the range of the function.

c As x approaches $-\infty$, what does y approach?



5 For each of the following functions:

i State the domain of the function.

ii State the range of the function.

iii Sketch the graph of the function.

a $y = -\sqrt{x} + 5$ **b** $y = -5\sqrt{x}$ **c** $y = -\sqrt{-x}$

6 Consider the function $y = 2\sqrt{x} + 3$.

a Is the function increasing or decreasing from left to right?

b Is the function more or less steep than $y = \sqrt{x}$?

c What are the coordinates of the vertex?

d Sketch the graph of the function.

7 Consider the function $y = \frac{1}{2}\sqrt{x} - 2$.

a Is the function increasing or decreasing from left to right?

b Is the function more or less steep than $y = \sqrt{x}$?

c What are the coordinates of the vertex?

d Sketch the graph of the function.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

8 Consider the function $y = -\frac{1}{2}\sqrt{x} + 2$.

- **a** Is the function increasing or decreasing from left to right?
- **b** Is the function more or less steep than $y = \sqrt{x}$?
- **c** What are the coordinates of the vertex?
- **d** Sketch the graph of the function.

9 Consider the function $y = -2\sqrt{x-2}$.

- **a** Is the function increasing or decreasing from left to right?
- **b** Is the function more or less steep than $y = \sqrt{x}$?
- **c** What are the coordinates of the vertex?
- **d** Sketch the graph of the function.

10 Consider the function $y = 3\sqrt{x-3} - 2$.

- **a** Is the function increasing or decreasing from left to right?
- **b** Is the function more or less steep than $y = \sqrt{x}$?
- **c** What are the coordinates of the vertex?
- **d** Sketch the graph of the function.

11 Consider the function $y = -\frac{1}{2}\sqrt{x-2} + 4$.

- **a** Is the function increasing or decreasing from left to right?
- **b** Is the function more or less steep than $y = -\sqrt{x}$?
- **c** What are the coordinates of the vertex?
- **d** Sketch the graph of the function.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

12 For each of the following functions:

i Sketch the graph of the function.

ii State the domain.

iii State the range.

a $f(x) = \sqrt{x} - 2$

b $f(x) = \sqrt{x+1}$

c $f(x) = \sqrt{-x} - 1$

d $f(x) = \sqrt{x+1} - 4$

e $f(x) = -\sqrt{x+1}$

f $f(x) = \sqrt{\dfrac{x-2}{4}}$

g $f(x) = -2\sqrt{x+5}$

h $f(x) = -\dfrac{\sqrt{x}}{2} - 2$

i $f(x) = \dfrac{\sqrt{x-1}}{2} + 2$

Characteristics of square root functions

13 Determine whether the following statements are true about the function $y = \sqrt{x}$.

a The function doesn't have any intercepts.

b The function has an asymptote.

c The function has a limiting value.

d The function is linear.

e As x increases, the function value approaches infinity.

14 What is the domain and range of $y = \sqrt{x} - 2$?

15 Consider the function $y = \sqrt{x-5}$.

a State the domain of the function.

b State the range of the function.

c Are the following statements true about the function $y = \sqrt{x-5}$?

i The functions $y = \sqrt{x}$ and $y = \sqrt{x-5}$ increase at the same rate.

ii The function $y = \sqrt{x-5}$ increases more rapidly than $y = \sqrt{x}$.

iii The function $y = \sqrt{x}$ increases more rapidly than $y = \sqrt{x-5}$.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

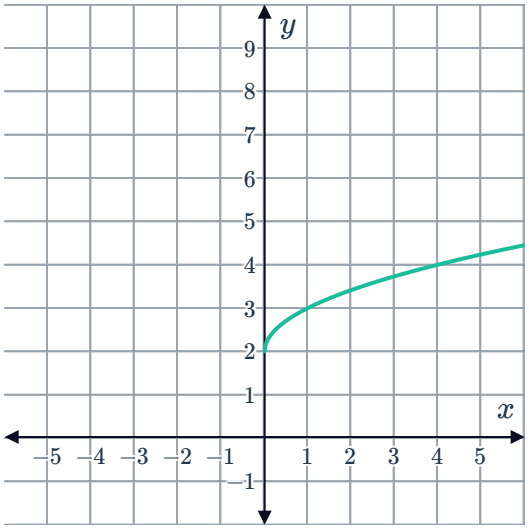
16 Consider the function $y = \sqrt{-x} + 6$.

- **a** What is the smallest possible function value?
- **b** State the domain of the function.
- **c** State the range of the function.

17 Consider the function $f(x) = \sqrt{x - 2} + 3$. State the domain of the function using interval notation.

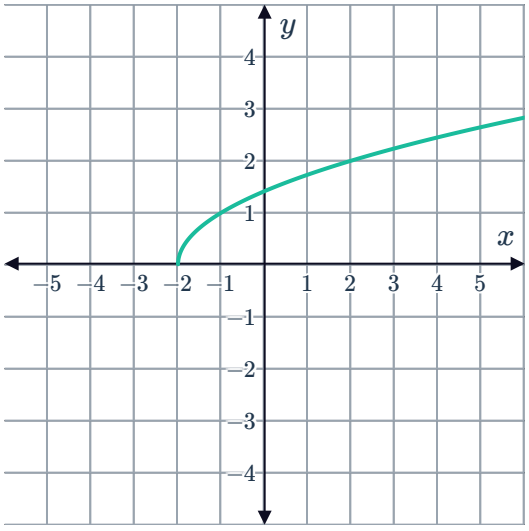
Transformations

18 The diagram shows the graph of a square root function $f(x)$:



- **a** What is the smallest y-value of the function?
- **b** Describe the transformation of $y = \sqrt{x}$ that has occurred to create the function $f(x)$.
- **c** State the equation of the function.

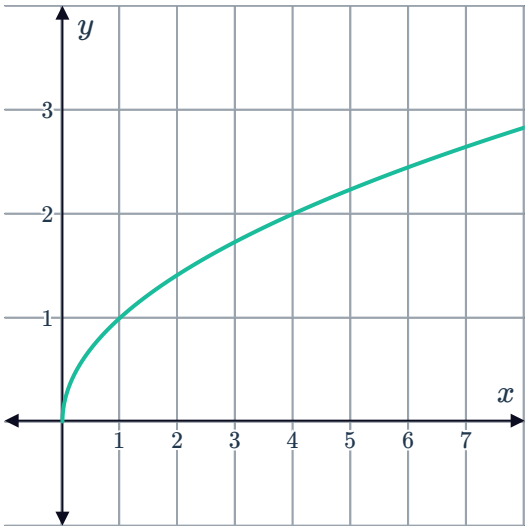
19 The diagram shows the graph of a square root function $f(x)$:



- **a** At what value of x is y equal to 0?
- **b** Describe the transformation of $y = \sqrt{x}$ that has occurred to create the function $f(x)$.
- **c** State the equation of the function.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

20 Consider the graph of $y = \sqrt{x}$ shown:



a

i Sketch the curve after $y = \sqrt{x}$ has been reflected across the y-axis.

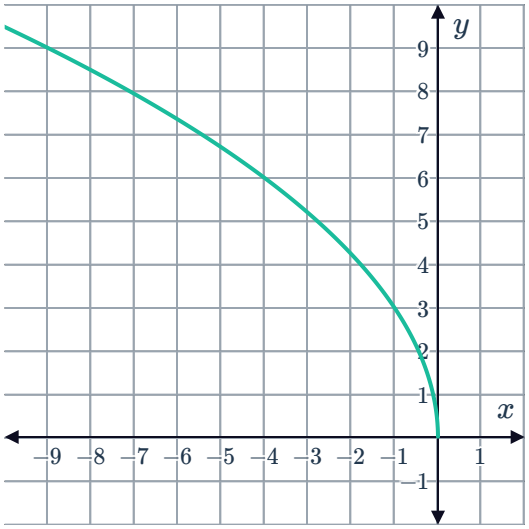
ii What is the equation of this new graph?

b

i Sketch the curve after $y = \sqrt{x}$ has been reflected across the x-axis.

ii What is the equation of this new graph?

21 Consider the graph of the square root function:



a What is the domain of the function?

b Find the equation of the square root function.

22 The graph of $y = \sqrt{x}$ has been translated to the graph of $y = -3 + \sqrt{x}$.

a What is the number of units that the graph of $y = \sqrt{x}$ has been translated vertically?

b What is the direction that the graph of $y = \sqrt{x}$ has been translated?

23 The graph of $y = \sqrt{x}$ has been translated to the graph of $y = \sqrt{x} - 4$.

a Describe the transformation that has occured on the original function.

b Sketch the graph of $y = \sqrt{x} - 4$.

24 Describe how the graph of $y = 3\sqrt{x}$ differs from the graph of $y = \sqrt{x}$.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

25 The graph of $y = \sqrt{x}$ has a vertex at $(0, 0)$. By considering the transformations that have taken place, state the coordinates of the vertex of the following functions:

a $y = \sqrt{x} - 2$ **b** $y = -\sqrt{x} + 3$

26 Consider the function $y = \sqrt{x - 1}$.

a Sketch the graph of the function.

b What transformation of the graph of $y = \sqrt{x}$ results in the graph of $y = \sqrt{x - 1}$?

27 Describe a series of transformations to $y = \sqrt{x}$ that would result in $y = -\sqrt{x} - 5$.

28 Consider the function $f(x) = \sqrt{x}$.

a Complete the table of values for each transformation of the function $f(x)$.

x	0	1	4	9	16
$y = f(x)$	0	1	2	3	4
$g(x) = 3f(x)$					
$h(x) = -4f(x)$					

b Graph $f(x)$ and $g(x)$ on the same coordinate plane.

c Describe how the graph of $f(x)$ can be transformed into the graph of $g(x)$.

d Graph $f(x)$ and $h(x)$ on the same coordinate plane.

e Describe how the graph of $f(x)$ can be transformed into the graph of $h(x)$.

29 Consider the function $y = \sqrt{x}$:

a Describe how we can transform the graph of $y = \sqrt{x}$ to get the graph of $y = \sqrt{x - 4} + 3$.

b Sketch the graph of $y = \sqrt{x - 4} + 3$.

30 For each of the following, determine the new equation for the transformed function:

a The graph of $y = \sqrt{x}$ is translated 5 units to the right.

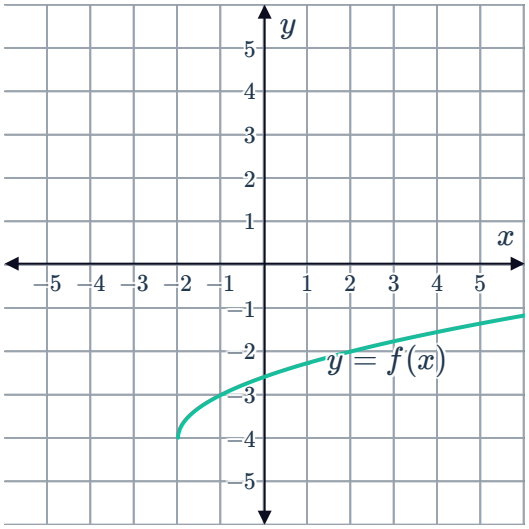
b The graph of $y = \sqrt{x}$ is reflected across the x-axis and then translated 6 units up.

c The graph of $y = \sqrt{x}$ is compressed vertically by a factor of $\frac{1}{2}$ and reflected across the x-axis.

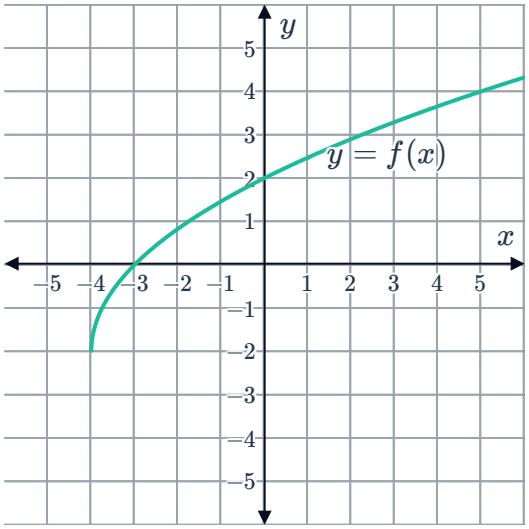
© ClipSAT: By Mr. Mohamed Abdallah +966597688647

31 Find the equation of the resulting function when $f(x) = \sqrt{x}$ is compressed horizontally by a factor of $\frac{1}{2}$ and compressed vertically by a factor of $\frac{1}{4}$.

32 Use the graph of $y = f(x)$ to graph $y = f(x - 3) + 4$.



33 Use the graph of $y = f(x)$ to graph $y = \frac{1}{2}f(x - 4) + 2$.



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

38 The length of a blue whale calf in its first few months is modeled approximately by the equation $l = 1.5\sqrt{t+4}$, where l represents its length in meters at t months of age.

a Complete the table of values:

Months (t)	0	60	96
Length (l)			

b Graph the length function.

c In this practical context, what are the possible values of t?

d Is the calf's length increasing by the same amount each month?

e According to the model, what will be the length of the whale at 7 years of age? Round your answer to one decimal place.

f An orca whale calf is born shorter than a blue whale calf. If its growth can be modeled by the equation $L = k\sqrt{t+4}$, what is the value of k?

39 The radius r of a sphere with surface area A is $r = \sqrt{\dfrac{A}{4\pi}}$.

Solve for the surface area, A, of a moon with a radius of 1080 kilometers. Round your answer to the nearest square kilometer.

40 A rectangle is such that its length is 2 times its width.
Let A represent the area of the rectangle and w represent its width.

a Form an equation for A in terms of w.

b Rearrange the equation to solve for w.

c Complete the table of values:

Area (A)	0	2	8	18	32
Width (w)					

d Graph the function $w = \sqrt{\dfrac{A}{2}}$.

e Using the graph, determine the area of the rectangle that has a width of 5.

f If w and A were the side length and area of a square respectively, the function would be $w = \sqrt{A}$. For $A \geq 1$, would the graph of this function be above or below the graph of $w = \sqrt{\dfrac{A}{2}}$?

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

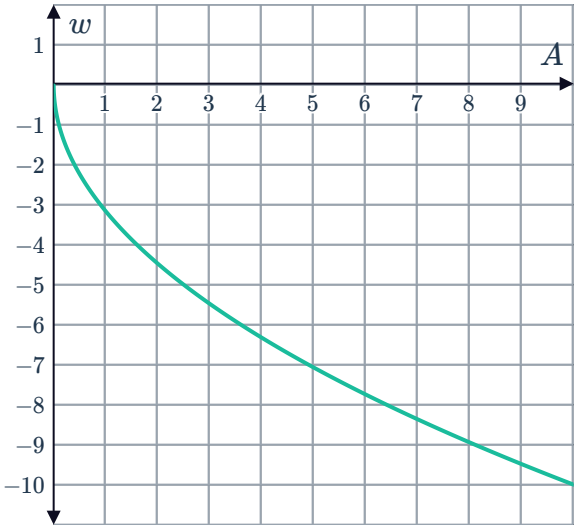
41 To model the rate at which an object sinks when it is dropped into water, a diver jumped from a height into a deep tank, and continued to allow herself to sink once she entered the water. Time was measured from when the diver hit the water, so the graph shows the diver's depth, D, at time t seconds after entering the water.

a Find the equation for D in terms of t.

b What was the diver's depth 5 seconds after entering the water? Express your answer in simplest radical form.

c What was the diver's average speed through the water in the first 5 seconds? (Note that speed is positive.)

d What was the diver's average speed through the water in the last 5 seconds (between $t = 5$ and $t = 10$)?



© ClipSAT: By Mr. Mohamed Abdallah +966597688647